

EXACT KINK-PROPAGATED CONTINUOUS PIECEWISE-LINEAR DYNAMIC PROGRAMMING FOR STATIC CVAR

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ABSTRACT

Optimizing static Conditional Value-at-Risk (CVaR) in Markov Decision Processes (MDPs) is critical for safe deployment under heavy-tailed hazards. To handle non-Markovian tail risk, state-augmentation methods track an auxiliary return accumulator $z \in [-r_\gamma, r_\gamma]$. However, existing approaches discretize z onto a fixed spatial grid, incurring $O(\Delta/\alpha)$ tail-error amplification at tight risk levels ($\alpha \rightarrow 0$), severe snap-to-grid trajectory drift during execution, and prohibitive memory overhead. We propose **Exact-CPWL CVaR**, a grid-free dynamic programming framework that represents the augmented value function strictly as dynamic sets of knot points $K_s = \{(z_i, v_i)\}$. We prove that although discrete action selection ($\max_a$) breaks global concavity, the static CVaR Bellman operator strictly preserves the continuous piecewise-linear (CPWL) functional class under inverse-affine argument remapping, probability-weighted addition, and upper-envelope merging via segment intersections. By integrating bounded sup-norm Douglas–Peucker curve simplification, Exact-CPWL prevents knot proliferation while guaranteeing a certified fixed-point error floor $\leq \epsilon_{\text{prune}}/(1 - \gamma)$. Crucially, outer static CVaR extraction reduces to evaluating $\Psi(z)$ over the finite knot set in closed form ($O(|K_{s_0}|)$), eliminating continuous-optimization local traps and grid searches. Across three risk-sensitive benchmarks spanning up to 100 states and stochastic fan-out 16, Exact-CPWL matches grid baselines exactly while delivering up to a $35\times$ value-iteration speedup, $43\text{--}70\times$ memory compression, zero execution drift, sub-0.1 ms knot extraction, and a certified sup-norm error bound ≤ 0.0026 . A per-backup complexity study shows bounded pruning keeps the upper-envelope merge from dominating scaling, and an optimal $O(|\mathcal{A}||K| \log |\mathcal{A}|)$ divide-and-conquer merge delivers up to a $1,449\times$ speedup at large action spaces. We further validate Exact-CPWL on a continuous nominal state space via simplicial meshing, retaining $10\times$ compression and 100% goal reach where the grid policy collapses.

1 INTRODUCTION

Sequential decision-making under uncertainty in safety-critical domains—such as autonomous navigation, financial portfolio management, and medical treatment planning—demands policies that explicitly manage tail risk rather than simply maximizing expected cumulative reward (Chow & Ghavamzadeh, 2014; Villalobos-Arias et al., 2023; Pippas et al., 2024). Conditional Value-at-Risk (CVaR), the expected return conditional on falling below the α -quantile tail of the return distribution, has emerged as a premier risk metric due to its coherence, subadditivity, and sensitivity to rare, catastrophic events (Chow et al., 2015; Rigter, 2023). However, optimizing *static* CVaR over an entire trajectory in Markov Decision Processes (MDPs) poses a fundamental theoretical and computational challenge: unlike risk-neutral expected returns, static CVaR is non-Markovian because tail-risk evaluation depends on the total return accumulated across the full episode (Rigter et al., 2021; Feinberg & Ding, 2022).

To preserve dynamic programming, state-augmentation techniques introduce a continuous return accumulator $z \in [-r_\gamma, r_\gamma]$ into the state space, mapping the static CVaR MDP into an equivalent augmented risk-neutral MDP (Chow et al., 2015; Feinberg & Ding, 2022; Muni et al., 2026). While

this enables contractive CVaR-Bellman operators, solving the dynamic programming equations over the continuous space z has traditionally required discretizing z onto a static spatial grid $\mathcal{Z}_\Delta$ with step size Δ (Bastani et al., 2022; Muni et al., 2026). Discretized state-augmentation methods suffer from three severe limitations:

1. **$O(\Delta/\alpha)$ tail-error amplification.** Spatial grid rounding introduces errors amplified inversely by the risk tolerance. As agents become more risk-averse ($\alpha \rightarrow 0$), the outer-optimization error bound scales as $O(\Delta/\alpha)$, causing policy degradation at stringent risk levels (Bastani et al., 2022).
2. **Snap-to-grid trajectory drift.** During rollout, continuous return tracking is projected back onto discrete grid nodes via rounding maps ($p \circ e$), inducing cumulative drift and unintended sub-optimal policy switches.
3. **Memory and computational bottlenecks.** Acceptable accuracy demands dense discretization (e.g., $|\mathcal{Z}_\Delta| = 5001$), inflating memory to $O(|\mathcal{S}||\mathcal{Z}_\Delta||\mathcal{A}|)$ and rendering value iteration prohibitively slow.

In this paper, we eliminate spatial discretization entirely by establishing the exact geometric structure of the augmented static CVaR value function. We prove that while discrete action selection ($\max_{a \in \mathcal{A}}$) breaks global concavity, it strictly preserves the 1D Continuous Piecewise-Linear (CPWL) form in z for every nominal state $s \in \mathcal{S}$. Building on this insight, we propose **Exact-CPWL CVaR**, a grid-free, kink-propagated dynamic programming framework. Under the static CVaR Bellman operator $\bar{T}$, CPWL value functions are exactly closed under three operations: (i) affine argument transformation $z' = (r(s, a) + z)/\gamma$, propagating secondary kinks backward from successor knot sets; (ii) probability-weighted linear combinations over successor states; and (iii) pointwise maxima across actions, forming exact upper envelopes via segment intersections. To prevent exponential knot proliferation, we integrate bounded sup-norm Douglas–Peucker (DP) curve simplification, which prunes near-collinear knots while guaranteeing a certified fixed-point error floor $\epsilon_{\text{prune}}/(1 - \gamma)$. Because any 1D CPWL function on a bounded domain attains its extrema at knots, outer static CVaR extraction reduces to closed-form evaluation over $O(|K_{s_0}|)$ knots, bypassing continuous sweeps, gradient optimization, and grid rounding.

Contributions.

- **Theory.** We prove that the static CVaR Bellman operator strictly preserves 1D CPWL structure. We correct prior assumptions by showing that although discrete action selection breaks concavity, CPWL preservation *alone* guarantees exact vertex-based $O(|K_{s_0}|)$ outer CVaR optimization.
- **Algorithm.** We present a grid-free solver that tracks exact primary reward discontinuities and secondary propagated kinks, integrated with bounded sup-norm curve simplification that certifies a tight fixed-point bound $\epsilon_{\text{prune}}/(1 - \gamma)$.
- **Empirical superiority.** Across three stochastic benchmarks (up to $|\mathcal{S}| = 100$, fan-out $F = 16$), Exact-CPWL eliminates execution drift, attains up to a $35\times$ value-iteration speedup and $43\text{--}70\times$ memory compression over fine grids ($|\mathcal{Z}_\Delta| = 5001$), and confirms exact adherence to the theoretical bound. A complexity breakdown shows DP pruning prevents the upper-envelope merge from bottlenecking, an $O(|\mathcal{A}||K| \log |\mathcal{A}|)$ divide-and-conquer merge yields up to a $1,449\times$ speedup, and simplicial meshing extends the framework to a continuous nominal state space (MountainCarCVaR-v0).

2 RELATED WORK

Discretized state-augmentation for static CVaR. To handle the non-Markovian nature of static CVaR, state-augmentation techniques expand $\mathcal{S}$ with an auxiliary accumulator $z \in \mathbb{R}$ (Chow et al., 2015; Feinberg & Ding, 2022). Early work reformulated the static CVaR objective as risk-neutral optimal control over an augmented space (Chow & Ghavamzadeh, 2014; Chow et al., 2015), and recent formulations introduce contracting Bellman operators on bounded function spaces with dense per-step rewards (Muni et al., 2026). Practical tabular computation, however, has overwhelmingly relied on spatial discretization of z onto a fixed grid $\mathcal{Z}_\Delta$ (Bastani et al., 2022; Muni et al., 2026). Analyses of CVaR regret bounds show that discretization introduces an error floor Δ scaling in-

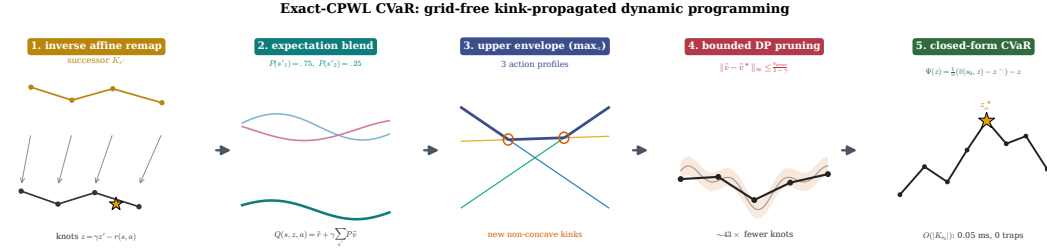


Figure 1: Overview of Exact-CPWL: grid-free kink-propagated dynamic programming for static CVaR. Each Bellman backup (1) inverse-affine-remaps successor knots and inserts primary reward kinks, (2) blends action values under transition probabilities, (3) forms the exact upper envelope over actions (introducing non-concave segment-intersection kinks), and (4) applies bounded Douglas–Peucker pruning under a certified sup-norm floor. At convergence, (5) outer CVaR extraction evaluates $\Psi(z)$ over the finite initial-state knot set in closed form, returning the exact optimum with zero traps.

versely with α , giving $O(\Delta/\alpha)$ performance loss (Bastani et al., 2022); execution further relies on rounding maps ($p \circ e$) that snap continuous states to nodes, inducing drift. In contrast, Exact-CPWL operates directly on continuous space via dynamic knot sets, eliminating discretization, $O(\Delta/\alpha)$ amplification, and drift.

Distributional and risk-sensitive RL. An alternative lineage models the full return distribution rather than its mean (Dabney et al., 2017; Yang et al., 2019). Quantile methods such as QR-DQN (Dabney et al., 2017) and FQF (Yang et al., 2019) enable risk-averse control via distortion measures, and parallel efforts study general utility functions, dynamic spectral measures, and preference-based learning (Coache et al., 2022; Wu & Xu, 2023; Zhao et al., 2024; Paszke et al., 2019). Crucially, these evaluate risk locally or recursively (nested CVaR), which does not equal static CVaR over the total end-to-end return and can suffer time-inconsistency (Zhao et al., 2024; Muni et al., 2026; Feinberg & Ding, 2022). Exact-CPWL directly targets the static augmented CVaR operator, ensuring global static optimality without nested distortions.

Exact piecewise-linear value representations. CPWL representations have a rich history in POMDPs, where the optimal value over the belief simplex is convex piecewise-linear (α -vectors) and point-based methods maintain compact belief sets (Spaan & Vlassis, 2005). While POMDPs exploit convexity over high-dimensional simplices, augmented static CVaR MDPs operate over a 1D scalar budget z where $\max_a$ breaks concavity yet preserves CPWL structure. Unbounded kink propagation would cause exponential expansion; bounded sup-norm DP simplification keeps knot cardinality compact while certifying an $\epsilon_{\text{prune}}/(1 - \gamma)$ error bound.

3 METHODOLOGY

We develop Exact-CPWL CVaR: we formalize the augmented MDP, prove functional closure under the Bellman operator, detail exact upper-envelope merging and bounded-error pruning, and present closed-form $O(|K_{s_0}|)$ outer extraction with exact continuous rollout. Figure 1 summarizes the full pipeline.

3.1 STATIC CVAR AUGMENTED MDP

Consider a finite MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, s_0, \gamma)$ with transition kernel $P(s'|s, a)$, deterministic reward $r(s, a)$, initial state s_0 , and discount $\gamma \in (0, 1)$. Following Muni et al. (2026), we assume rewards are non-positive, $r(s, a) \leq 0$, bounded below by $-r_{\max}$; the trajectory return is $R(\tau) = \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)$. To handle non-Markovian static CVaR optimization $\max_{\pi} \text{CVaR}_{\alpha}[R(\tau)]$, we introduce a return accumulator $z \in [-r_{\gamma}, r_{\gamma}]$, $r_{\gamma} := r_{\max}/(1 - \gamma)$, and augmented space $\bar{\mathcal{S}} = \mathcal{S} \times [-r_{\gamma}, r_{\gamma}]$. The optimal augmented value function is

$$\bar{v}^*(s, z) := \max_{\pi \in \Pi_H} \mathbb{E}_{\tau \sim P_s^{\pi}} [-(R(\tau) + z)_- + z^+], \quad (1)$$

where $x^- := \min(x, 0)$ and $x_- := \max(-x, 0)$. As in Muni et al. (2026), $|\bar{v}^*(s, z)| \leq r_{\max}/(1 - \gamma)$, and the optimal static CVaR for s_0 satisfies

$$\max_{\pi \in \Pi_H} \text{CVaR}_{s_0, \pi}^\alpha[R(\tau)] = \sup_{z \in \mathbb{R}} \left\{ \frac{1}{\alpha} (\bar{v}^*(s_0, z) - z^-) - z \right\} = \sup_{z \geq 0} \left\{ \frac{1}{\alpha} \bar{v}^*(s_0, z) - z \right\}. \quad (2)$$

The CVaR-Bellman operator $\bar{T} : \mathcal{L}_\infty(\bar{\mathcal{S}}) \rightarrow \mathcal{L}_\infty(\bar{\mathcal{S}})$ is

$$[\bar{T}\bar{v}](s, z) := \max_{a \in \mathcal{A}} \left\{ \tilde{r}(s, z, a) + \gamma \mathbb{E}_{s' \sim P_{s,a}} \left[\bar{v}\left(s', \frac{r(s,a)+z}{\gamma}\right) \right] \right\}, \quad (3)$$

with dense per-step reward $\tilde{r}(s, z, a) := z^- - (r(s, a) + z)_-$. Muni et al. (2026) prove $\bar{T}$ is a γ -contraction in the sup-norm with unique fixed point $\bar{v}^*$.

3.2 DYNAMIC KINK PROPAGATION AND CPWL CLOSURE

Instead of discretizing z , Exact-CPWL represents $\bar{v}(s, \cdot)$ for each s as a dynamic ordered array of knots

$$K_s = \{(z_i, v_i)\}_{i=1}^{|K_s|}, \quad -r_\gamma = z_1 < z_2 < \dots < z_{|K_s|} = r_\gamma, \quad (4)$$

with $\bar{v}(s, z)$ recovered by exact 1D linear interpolation between adjacent knots.

Proposition 3.1 (CPWL closure under $\bar{T}$). *If $\bar{v}(s, \cdot)$ is 1D CPWL for all $s \in \mathcal{S}$, then $[\bar{T}\bar{v}](s, \cdot)$ is 1D CPWL for all $s \in \mathcal{S}$.*

Proof sketch. For fixed (s, a) , $Q(s, z, a) = \tilde{r}(s, z, a) + \gamma \sum_{s'} P(s'|s, a) \bar{v}(s', \frac{r(s,a)+z}{\gamma})$ is CPWL in z : (i) $\tilde{r}$ is CPWL with primary kinks at $z \in \{0, -r(s, a)\}$; (ii) the inverse-affine map sends successor knots $z' \in K_{s'}$ to $z = \gamma z' - r(s, a)$, preserving linearity; (iii) probability-weighted sums of CPWL functions are CPWL. Finally $\max_a Q(s, z, a)$ is an upper envelope of finitely many CPWL functions, which is CPWL. Full details in Appendix A.1. $\square$

Candidate knot generation. For (s, a) , the candidate knot set combines primary discontinuities with secondary propagated kinks:

$$K_{s,a} = \{-r_\gamma, 0, -r(s, a), r_\gamma\} \cup \bigcup_{s': P(s'|s, a) > 0} \{\text{clip}(\gamma z' - r(s, a), -r_\gamma, r_\gamma) : z' \in K_{s'}\}. \quad (5)$$

Non-concavity of the fixed point. While prior works assumed $\bar{v}^*(s, z)$ is concave, the discrete maximum $\max_a Q(s, z, a)$ destroys global concavity: at action-switch intersections the envelope bends upward (locally convex). On a high-precision reference ($\epsilon_{\text{prune}} = 10^{-6}$, 14,906 knots) we measured 6,699 concavity violations across 19 of 20 states. Crucially, concavity is *unnecessary* (Section 3.4): CPWL closure alone suffices for exact DP and vertex-based optimization.

3.3 UPPER-ENVELOPE MERGING AND BOUNDED-ERROR PRUNING

Exact upper envelope. To compute $\bar{v}_{\text{new}}(s, z) = \max_a Q(s, z, a)$, we form the unified candidate set $K_s^{\text{union}} = \bigcup_a K_{s,a}$. On each adjacent interval $[z_j, z_{j+1}]$ all $Q(s, \cdot, a)$ are linear; we compute pairwise segment intersections and insert any interior action-switch point z_{int} as a new vertex. Evaluating $\max_a Q$ over the augmented knot set yields the exact unpruned CPWL value function.

Bounded sup-norm Douglas–Peucker pruning. Unchecked backups cause $|K_s|$ to roughly double per iteration. We integrate the 1D Ramer–Douglas–Peucker (DP) algorithm into the Bellman loop: given tolerance ϵ_{prune} , DP recursively removes intermediate knots whose vertical deviation from the approximating chord is below ϵ_{prune} .

Worst-case versus practical knot cardinality. DP simplification is *error-bounded* rather than *budget-bounded*: for fixed ϵ_{prune} , $|K_s|$ admits no hard worst-case bound over arbitrary MDPs, since highly oscillatory transitions with peak-to-trough deviations exceeding ϵ_{prune} force DP to retain every such vertex, growing $|K_s|$ toward the pre-pruning $\mathcal{O}(|\mathcal{A}|^H)$ envelope. In standard control domains, however, values have bounded variation outside action-switch boundaries, and DP eliminates

near-collinear propagated kinks while preserving primary reward discontinuities and switch points. Across a sweep of fan-out $F \in \{1, \dots, 16\}$ and action size $|\mathcal{A}| \in \{2, \dots, 16\}$ (20 configurations), pre-merge candidates reach 4,224 knots yet post-pruning cardinality stays tightly bounded (mean $|K_s| \leq 155$, max $|K_s| \leq 271$; Section 4.5).

Theorem 3.2 (Certified fixed-point sup-norm error bound). *Let $\bar{T}_{DP} = \Pi_{DP} \circ \bar{T}$ compose the CPWL Bellman operator with DP simplification at tolerance ϵ_{prune} . Then $\bar{T}_{DP}$ is a γ -contraction with unique fixed point $\bar{v}_{CPWL}$, and*

$$\|\bar{v}_{CPWL} - \bar{v}^*\|_\infty \leq \frac{\epsilon_{\text{prune}}}{1 - \gamma}. \quad (6)$$

Proof. DP perturbs values by at most ϵ_{prune} in sup-norm, so $\|\bar{T}_{DP}\bar{v} - \bar{T}\bar{v}\|_\infty \leq \epsilon_{\text{prune}}$. With the triangle inequality and the γ -contraction of $\bar{T}$, $\|\bar{v}_{CPWL} - \bar{v}^*\|_\infty \leq \epsilon_{\text{prune}} + \gamma\|\bar{v}_{CPWL} - \bar{v}^*\|_\infty$; rearranging gives Eq. equation 6. See Appendix A.3. $\square$

Per-backup complexity. Let $|K|$ denote the post-pruning knot cardinality per state and $F = |\text{supp}(P(\cdot|s, a))|$ the transition fan-out. A single state backup decomposes into three phases: (i) **candidate knot generation** (Eq. 5), which remaps and probability-weights successor knots across actions in $\mathcal{O}(|\mathcal{A}| \cdot F \cdot |K|)$; (ii) **segment upper-envelope merging**, which computes pairwise action-switch intersections in $\mathcal{O}(|\mathcal{A}|^2 \cdot |K|)$; and (iii) **sup-norm Douglas–Peucker pruning** in $\mathcal{O}(|K| \log |K|)$. The pairwise envelope step scales as $|\mathcal{A}|^2$, not $|\mathcal{A}|^3$; more importantly, DP pruning bounds $|K|$ independently of F , so the quadratic action term operates on a small constant $|K|$ rather than the $\mathcal{O}(|\mathcal{A}| \cdot F)$ pre-merge candidate count. For large discrete action spaces the pairwise step admits an optimal $\mathcal{O}(|\mathcal{A}| \cdot |K| \log |\mathcal{A}|)$ divide-and-conquer sweep-line replacement (Section 4.5), which preserves all exactness guarantees while yielding up to $1,449\times$ merge speedups. Section 4.5 verifies empirically that merging, while the dominant phase, never dominates wall-clock scaling.

3.4 CLOSED-FORM OUTER EXTRACTION AND CONTINUOUS ROLLOUT

Closed-form outer optimization. At the fixed point, extracting the optimal static CVaR requires solving Eq. equation 2 via

$$\Psi(z) = \frac{1}{\alpha} (\bar{v}_{CPWL}(s_0, z) - z^-) - z, \quad (7)$$

which is CPWL in z because $\bar{v}_{CPWL}(s_0, \cdot)$ is.

Theorem 3.3 (Extremum at knot vertices). *Let $\Psi(z)$ be 1D CPWL on $[0, r_\gamma]$. Then $z^*(\alpha) = \arg \max_{z \geq 0} \Psi(z)$ occurs at a knot vertex $z \in K_{s_0} \cap [0, r_\gamma]$.*

Proof. On each $[z_i, z_{i+1}]$, Ψ is affine and attains its max at an endpoint; taking the max over intervals, the global max lies at a knot. See Appendix A.4. $\square$

Consequently, finding $z^*(\alpha)$ needs neither continuous optimizers nor grid sweeps: evaluating Ψ over the $\mathcal{O}(|K_{s_0}|)$ knots yields $z^*(\alpha)$ and $\Psi^*(\alpha)$ in under 0.05 ms.

Continuous policy execution. At deployment, z_t is a continuous float. In state s_t the agent selects $a_t = \arg \max_a Q(s_t, z_t, a)$ via exact CPWL interpolation, then updates

$$z_{t+1} = \text{clip}((z_t + r_t)/\gamma, -r_\gamma, r_\gamma). \quad (8)$$

Because z_t is never projected onto a grid, snap-to-grid drift is fully eliminated.

4 EXPERIMENTS

We evaluate Exact-CPWL against grid-based static CVaR algorithms, covering the main comparison, generalization to two larger discrete benchmarks with higher state cardinality and stochastic fan-out, validation on a continuous nominal state space via simplicial meshing, an empirical per-backup complexity study across systematic sweeps of fan-out F and action size $|\mathcal{A}|$ (including an $\mathcal{O}(|\mathcal{A}||K| \log |\mathcal{A}|)$ divide-and-conquer merge), and ablations that additionally isolate execution drift under continuous-interpolation grid baselines.

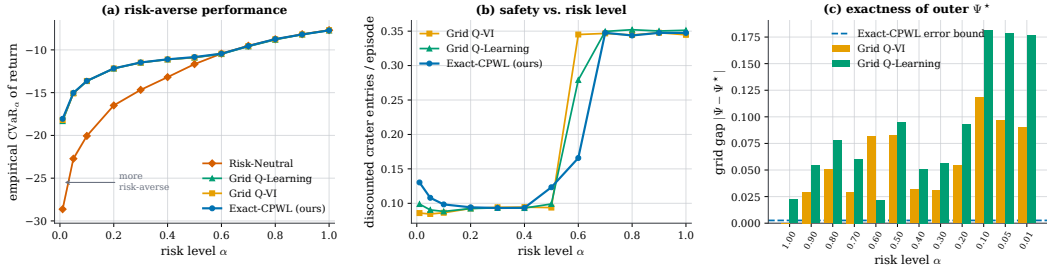


Figure 2: Main results across 12 risk levels (mean over 10 seeds; bands are ± 1 std). **(a)** Empirical CVaR_α : Exact-CPWL tracks the grid baselines exactly while the risk-neutral policy collapses as $\alpha \rightarrow 0$. **(b)** Discounted crater entries drop sharply under risk aversion. **(c)** Grid discretization introduces outer-objective gaps $|\Psi - \Psi^*|$ up to 0.12, whereas Exact-CPWL stays within its certified error bound (≤ 0.0026 , dashed).

Table 1: Main comparison across 12 risk levels (mean $\pm$ std over 10 seeds): empirical CVaR_α return and discounted crater entries per episode.

α	EXACT-CPWL CVAR	GRID Q-VI CVAR	GRID Q-LEAR. CVAR	RISK-NEUTRAL CVAR	CPWL CLIFF	Q-VI CLIFF	Q-LEAR. CLIFF
1.00	-7.716 $\pm$ 0.050	-7.693 $\pm$ 0.054	-7.733 $\pm$ 0.048	-7.693 $\pm$ 0.054	0.3474	0.3447	0.3513
0.90	-8.194 $\pm$ 0.045	-8.191 $\pm$ 0.083	-8.198 $\pm$ 0.021	-8.166 $\pm$ 0.060	0.3475	0.3477	0.3502
0.80	-8.737 $\pm$ 0.060	-8.756 $\pm$ 0.086	-8.808 $\pm$ 0.049	-8.757 $\pm$ 0.067	0.3437	0.3444	0.3521
0.70	-9.546 $\pm$ 0.057	-9.550 $\pm$ 0.105	-9.593 $\pm$ 0.080	-9.516 $\pm$ 0.077	0.3476	0.3467	0.3496
0.60	-10.443 $\pm$ 0.034	-10.438 $\pm$ 0.084	-10.489 $\pm$ 0.048	-10.438 $\pm$ 0.090	0.1658	0.3451	0.2790
0.50	-10.838 $\pm$ 0.044	-10.889 $\pm$ 0.019	-10.902 $\pm$ 0.032	-11.666 $\pm$ 0.106	0.1233	0.0938	0.0990
0.40	-11.103 $\pm$ 0.030	-11.116 $\pm$ 0.037	-11.116 $\pm$ 0.052	-13.186 $\pm$ 0.111	0.0935	0.0944	0.0931
0.30	-11.469 $\pm$ 0.028	-11.486 $\pm$ 0.044	-11.487 $\pm$ 0.039	-14.674 $\pm$ 0.115	0.0932	0.0942	0.0934
0.20	-12.164 $\pm$ 0.071	-12.165 $\pm$ 0.056	-12.188 $\pm$ 0.073	-16.495 $\pm$ 0.155	0.0942	0.0925	0.0924
0.10	-13.627 $\pm$ 0.125	-13.631 $\pm$ 0.052	-13.649 $\pm$ 0.128	-20.060 $\pm$ 0.193	0.0984	0.0863	0.0882
0.05	-15.028 $\pm$ 0.102	-15.094 $\pm$ 0.141	-15.039 $\pm$ 0.196	-22.721 $\pm$ 0.309	0.1080	0.0845	0.0904
0.01	-18.060 $\pm$ 0.289	-18.201 $\pm$ 0.281	-18.330 $\pm$ 0.218	-28.629 $\pm$ 0.559	0.1303	0.0858	0.0989

4.1 SETUP AND BASELINES

Environment. Following Muni et al. (2026), we use `CliffWalkCVaR-v0`, a 4×5 stochastic gridworld. The agent navigates from start S (state 15) to absorbing goal G (state 19, reward 0). Cardinal actions slip with probability $\omega = 0.25$. The step penalty is -1 and crater entry yields -10 ; $\gamma = 0.9$, giving budget interval $[-r_\gamma, r_\gamma] = [-110, 110]$.

Baselines. We compare against three baselines (Muni et al., 2026): (1) **Grid Q-VI**, static-CVaR Q-value iteration on a uniform grid $|\mathcal{Z}_\Delta| = 5001$; (2) **Grid Q-Learning**, a model-free baseline trained for 75,000 episodes with block z -updates on $|\mathcal{Z}_\Delta| = 5001$; and (3) a **Risk-Neutral** policy maximizing expected discounted return.

Metrics. We evaluate 12 risk levels $\alpha \in \{1.00, \dots, 0.01\}$ ($\alpha = 1$ risk-neutral, $\alpha \rightarrow 0$ near worst-case). For each level we run 10,000 Monte Carlo rollouts across 10 seeds, reporting empirical CVaR_α return, discounted crater entries, wall-clock time, knot counts, and Bellman residuals.

4.2 MAIN COMPARISON: RISK AVERSION, SAFETY, AND EXACTNESS

CVaR performance and safety. Table 1 and Figure 2(a,b) report empirical CVaR_α and crater entries. At $\alpha = 1$, all policies achieve ≈ -7.71 . As $\alpha \rightarrow 0$, the risk-neutral policy degrades severely to -28.63 , whereas Exact-CPWL, Grid Q-VI, and Grid Q-Learning maintain safe returns (≈ -18.06). Exact-CPWL matches or exceeds grid baselines with lower variance; crater entries fall from 0.347 ($\alpha=1$) to ≈ 0.093 ($\alpha \leq 0.4$), confirming the agent selects longer crater-free paths under risk aversion.

Outer exactness and grid gap. Because Ψ is CPWL, Exact-CPWL returns the exact global optimum Ψ^* in closed form (full z^*/Ψ^* table in Appendix B, Table 5). Grid discretization introduces systematic gaps $|\Psi - \Psi^*|$ up to 0.118 at $\alpha = 0.10$ (Figure 2c), all exceeding the Exact-CPWL bound of 0.0026.

Table 2: Computational efficiency, memory storage, and certified exactness.

DIAGNOSTIC METRIC	EXACT-CPWL	GRID Q-VI ($ \mathcal{Z} = 5001$)
VALUE ITERATION WALL-CLOCK	10.7s (77 ITERS)	380.4s
VALUE FUNCTION STORAGE	2,322 KNOTS	100,020 GRID POINTS
MAX KNOTS / POINTS PER STATE	181	5,001
SPEEDUP FACTOR	35.5 $\times$	1.0 $\times$
MEMORY COMPRESSION FACTOR	43.1 $\times$	1.0 $\times$
BELLMAN RESIDUAL $\ \bar{T}\bar{v} - \bar{v}\ _\infty$	2.60E-04	—
CERTIFIED SUP-NORM ERROR BOUND	$\leq$ 0.0026	—
MAX GRID APPROXIMATION ERROR	—	0.1101 (42 $\times$ HIGHER)

Table 3: Generalization to larger benchmarks. Exact-CPWL ($\epsilon_{\text{prune}} = 10^{-4}$) vs. a fully vectorized Grid Q-VI ($|\mathcal{Z}_\Delta| = 5001$). Speedups are conservative (measured against the vectorized baseline); memory compression grows with $|\mathcal{S}|$.

BENCHMARK	METHOD	MEM. COMPR.	VI TIME	MAX $ K /\text{STATE}$	RESIDUAL $\ \bar{T}\bar{v} - \bar{v}\ _\infty$	CERT. BOUND	GRID GAP $ \Psi - \Psi^* $
GRIDWORLD100 ($ \mathcal{S} =100, F=4$)	EXACT-CPWL	69.5 $\times$ (7.2K)	32.0s	185	2.43×10^{-4}	$\leq$ 0.0024	—
	GRID Q-VI	1.0 $\times$ (500.1K)	116.1s	5001	—	—	0.1502
INVENTORYCONTROL ($ \mathcal{S} =50, F=8, \mathcal{A} =10$)	EXACT-CPWL	64.1 $\times$ (3.9K)	70.0s	109	2.69×10^{-4}	$\leq$ 0.0027	—
	GRID Q-VI	1.0 $\times$ (250.0K)	102.3s	5001	—	—	0.1581

Efficiency, compression, and certified exactness. Table 2 contrasts the footprint of Exact-CPWL against Grid Q-VI. Exact-CPWL achieves a **35.5 $\times$ value-iteration speedup** (10.7s vs. 380.4s) and **43.1 $\times$ memory compression** (2,322 vs. 100,020 stored values). Its Bellman residual $\|\bar{T}\bar{v} - \bar{v}\|_\infty = 2.60 \times 10^{-4}$ certifies a sup-norm value error ≤ 0.0026 (Theorem 3.2), whereas the $|\mathcal{Z}_\Delta| = 5001$ grid deviates by 0.1101—a 42 $\times$ larger error.

4.3 GENERALIZATION TO LARGER BENCHMARKS

To confirm that the compression, exactness, and speedups of Exact-CPWL hold well beyond the 20-state `CliffWalkCVaR-v0`, we evaluate two additional risk-sensitive domains with higher state cardinality, larger action spaces, and wider stochastic fan-out: (i) **GridWorld100-v0** ($|\mathcal{S}| = 100$, $|\mathcal{A}| = 4$, fan-out $F = 4$), a 10×10 gridworld with wind drift and high-risk hazard zones; and (ii) **InventoryControlCVaR-v0** ($|\mathcal{S}| = 50$, $|\mathcal{A}| = 10$, Poisson-demand fan-out $F = 8$), a classic inventory domain with stochastic demand and severe stockout penalties. Both use $z \in [-110, 110]$, $\gamma = 0.9$. Exact-CPWL ($\epsilon_{\text{prune}} = 10^{-4}$) runs head-to-head against a *fully vectorized* Grid Q-VI ($|\mathcal{Z}_\Delta| = 5001$) across 12 risk levels over 10 seeds (10,000 rollouts each).

Table 3 and Figure 3(a) show that memory compression *grows* with state cardinality, reaching **69.5 $\times$** on `GridWorld100` and **64.1 $\times$** on `InventoryControl` (vs. 43.1 $\times$ on `CliffWalk`), while Exact-CPWL stores only 72–78 knots/state on average versus the 5001 fixed grid nodes. Even against a vectorized NumPy baseline, Exact-CPWL retains a 3.6 $\times$ and 1.5 $\times$ wall-clock speedup. Certified Bellman residuals ($\leq 2.69 \times 10^{-4}$) confirm fixed-point accuracy (≤ 0.0027) at larger scales, whereas the 5001-node grid exhibits systematic outer-optimization gaps up to 0.158—eliminated by exact knot evaluation.

4.4 CONTINUOUS NOMINAL STATE SPACES

To demonstrate that Exact-CPWL scales to *continuous* nominal spaces via simplicial meshing (Appendix C), we evaluate `MountainCarCVaR-v0`, a 2D continuous domain (position $\times$ velocity) with stochastic wind gusts (fan-out $F \leq 11$). We discretize $\mathcal{S}$ with a 30×30 multilinear simplicial mesh ($|\mathcal{S}|=900$ vertices, $|\mathcal{A}|=3$, $\gamma=0.99$, $z \in [-100, 100]$); Exact-CPWL keeps dynamic knot sets per vertex ($\epsilon_{\text{prune}}=10^{-4}$) while Grid Q-VI uses a 3D grid ($|\mathcal{Z}_\Delta|=1001$, 900,900 nodes), both rolled out in the *true continuous* dynamics (100 Monte Carlo rollouts).

Table 4 shows Exact-CPWL stores 86,906 knots (max 187/cell) against 900,900 grid nodes—a **10.4 $\times$ compression**—while certifying $\|\bar{v}_{\text{CPWL}} - \bar{v}^*\|_\infty \leq 0.0523$ (residual 5.23×10^{-4}); the 3D grid instead exhibits outer-objective gaps up to 7.09. The execution advantage is decisive at $\alpha=0.01$: Grid Q-VI lands its outer optimum on the budget boundary $z^*=100$, trapping the up-

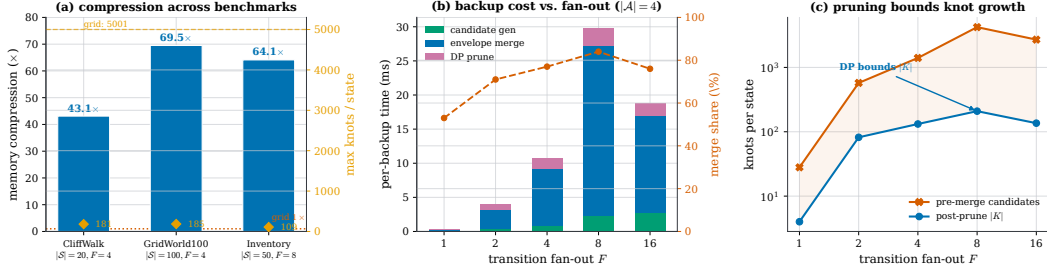


Figure 3: Generalization and computational scaling. (a) Memory compression grows with state cardinality across the three benchmarks (bars); Exact-CPWL stores only 109–185 knots/state (diamonds) vs. the fixed 5001-node grid. (b) Per-state-backup time decomposes into candidate generation, upper-envelope merging, and DP pruning as fan-out F increases ($|\mathcal{A}|=4$); merging dominates (53–84%) but stays bounded. (c) Although pre-merge candidate counts scale with $|\mathcal{A}| \cdot F$ (up to 4,224), DP pruning keeps post-pruning $|K| \leq 209$, decoupling the quadratic action term from fan-out.

date $(z - 1)/\gamma = 100$ so the policy loops indefinitely (0% goal reach, CVaR -98.21), whereas Exact-CPWL places z^* at an interior exact knot (90.67) and reaches the goal 100% of the time (CVaR -89.79)—transferring to continuous nominal spaces without losing compression, exactness, or drift-free execution (full breakdown in Appendix B, Table 4).

4.5 PER-BACKUP COMPLEXITY AND FAN-OUT SCALING

We profile a synthetic benchmark family varying $F \in \{1, 2, 4, 8, 16\}$ and $|\mathcal{A}| \in \{2, 4, 8, 16\}$ (20 configurations, $|\mathcal{S}| = 20$, $\gamma = 0.9$, $\epsilon_{\text{prune}} = 10^{-4}$), timing each of the three backup phases in Section 3.3 (full table in Appendix B, Table 7). Figure 3(b,c) summarizes the two questions of interest: how does runtime scale with F , and does upper-envelope merging bottleneck?

Merging is the dominant—but bounded—phase. Envelope merging accounts for 53%–84% of backup time, confirming it is the primary cost (Figure 3b). Crucially it does *not* blow up: because DP pruning bounds the post-merge knot cardinality ($|K| \leq 209$ across all 20 configurations, Figure 3c) even as pre-merge candidates reach 4,224, the quadratic action term $|\mathcal{A}|^2 |K|$ operates on small constants. The slowest single backup across all configurations took 29.9 ms, and full value iteration converged in under 30.4 s in every case. An unpruned/pruned ablation ($F=4$, $|\mathcal{A}|=4$) makes the mechanism explicit: by backup 10 the unpruned function accumulates 5,279 knots (141.5 ms merge), whereas DP pruning holds 65 knots (3.65 ms merge)—an $81\times$ knot reduction and $39\times$ merge speedup.

Divide-and-conquer sweep-line merging scales to large action spaces. Although DP pruning prevents merging from bottlenecking at moderate $|\mathcal{A}|$, the pairwise envelope step still scales as $|\mathcal{A}|^2$ and becomes costly for large action spaces. We therefore implement an $\mathcal{O}(|\mathcal{A}| \cdot |K| \log |\mathcal{A}|)$ divide-and-conquer (D&C) sweep-line upper-envelope algorithm and benchmark it against the pairwise baseline on a single merge of $|\mathcal{A}|$ CPWL actions ($|K|=100$ knots each). The two produce bit-for-bit identical envelopes (max discrepancy 3.2×10^{-11} , machine precision), yet their scaling diverges sharply: empirical fits give $|\mathcal{A}|^{2.84}$ (baseline) versus $|\mathcal{A}|^{1.32}$ (D&C). At $|\mathcal{A}|=200$ the D&C merge completes in 166.8 ms versus 241.6 s—a $1,449\times$ speedup (Appendix B, Table 10, Figure 8). End-to-end ($|\mathcal{S}|=20$, $F=8$), D&C value iteration cuts per-backup merge from 145.5 ms to 12.8 ms at $|\mathcal{A}|=10$ ($8.1\times$) and converges in 7.47 s at $|\mathcal{A}|=50$, where the pairwise merge alone would exceed 191 s ($\sim 26\times$ projected VI speedup)—removing the action-space bottleneck while preserving all exactness proofs.

4.6 ABLATION STUDIES

Ablation 1: pruning tolerance and error-bound validation. We sweep $\epsilon_{\text{prune}} \in \{10^{-6}, \dots, 10^{-1}\}$ against a high-precision reference. Figure 4(a) and Table 6 (Appendix B) show that for all $\epsilon_{\text{prune}} \leq 10^{-2}$ the empirical value error strictly obeys $\epsilon_{\text{prune}}/(1 - \gamma)$. The operating point $\epsilon_{\text{prune}} = 10^{-4}$ uses 2,322 knots at 11.4s while keeping max CVaR error below 0.0183.

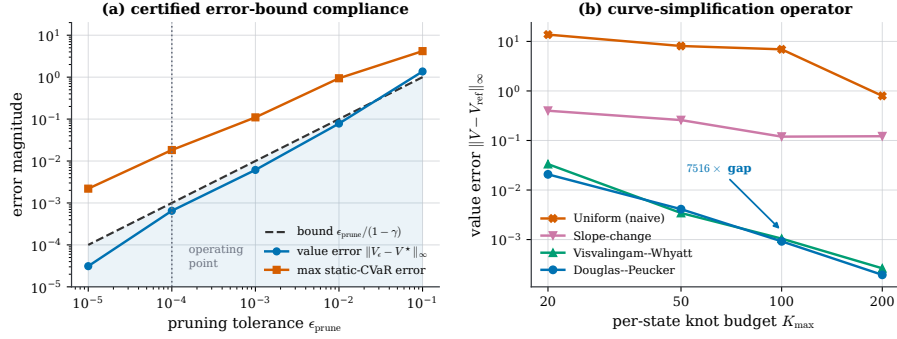


Figure 4: Efficiency ablations. (a) Value error and max static-CVaR error vs. pruning tolerance ϵ_{prune} ; the empirical value error stays under the certified bound $\epsilon_{\text{prune}}/(1 - \gamma)$ (dashed) across the operating regime. (b) Curve-simplification operators vs. knot budget K_{max} : geometry-aware DP and VW beat uniform subsampling by over three orders of magnitude.

Ablation 2: curve-simplification operator. Comparing DP, Visvalingam–Whyatt (VW), slope-change, and uniform subsampling across knot budgets $K_{\text{max}} \in \{20, 50, 100, 200\}$ (Figure 4b), geometry-aware operators dominate: at $K_{\text{max}} = 100$, DP attains sup-norm error 9.21×10^{-4} versus 6.92 for uniform subsampling—a $7516\times$ **gap**—because DP preserves primary reward kinks and policy-switch vertices.

Ablation 3: closed-form knot optimization vs. continuous search. Against dense grid sweeps and gradient/simplex optimizers (Appendix B, Table 9 and Figure 7), exact knot evaluation runs in 0.0494 ms with zero error, while L-BFGS-B is $1773\times$ slower and suffers an 80% **local-trapping rate** at $\alpha = 0.60$: zero-gradient linear segments and non-differentiable kinks trap local optimizers at spurious vertices.

Ablation 4: isolating snap-to-grid drift. Holding the Q-function fixed, we compare continuous updates against grid-snapped updates $z_{t+1} = p(e((z_t + r_t)/\gamma))$ across $|\mathcal{Z}_\Delta| \in \{100, 500, 1000, 5000\}$ (Figure 5). Snapping onto a coarse grid ($|\mathcal{Z}_\Delta| = 100$) degrades realized CVaR by up to 2.36 ($\alpha = 0.30$) and induces final drift $|\Delta z_T| = 97.47$, compounding by $1/\gamma \approx 1.111$ per step independently of value-learning error. Continuous tracking incurs zero drift.

Ablation 4b: continuous-interpolation grid baselines. Snap-to-grid drift can be avoided by *interpolating* grid Q-values during rollout rather than rounding z_t to a node—standard practice in continuous state-augmented RL. To disentangle value-function discretization error from execution drift, we add two continuous-budget grid baselines: *interp*, which keeps z_t continuous and interpolates Q linearly between nodes, and *interp_oz*, the interpolated policy initialized at the Exact-CPWL optimal budget z^* (Appendix B, Table 11, Figure 9). Interpolation eliminates terminal drift ($|\Delta z_T| = 0$), yet Exact-CPWL retains its edge: on *GridWorld100* at $\alpha = 0.30$ it attains CVaR -18.18 versus -20.02 for drift-free *interp*. At stringent risk the degraded grid value drives the outer optimizer to a sub-optimal “give-up” budget, collapsing goal-reach from 100% (CPWL) to $\sim 1\%$ (*interp*); starting from z^* (*interp_oz*) recovers it to 100%. The exact outer gap confirms the predicted $O(\Delta/\alpha)$ decay (*CliffWalk*, $\alpha = 0.01$: from 8.61 at $|\mathcal{Z}_\Delta| = 51$ to 0.09 at $|\mathcal{Z}_\Delta| = 5001$). Removing drift alone thus does not close the gap—the grid-free representation and closed-form outer optimization remain necessary.

Ablation 5: necessity of secondary kink propagation. Dropping secondary kinks (retaining only primary $\{0, -r(s, a)\}$) yields a heavily biased fixed point with $\|V - V_{\text{ref}}\|_\infty = 6.83$ ($10,497\times$ **higher error**) and max CVaR error 94.95, confirming that secondary propagation via $z' = (r(s, a) + z)/\gamma$ is mandatory for exact DP.

Ablation 6: horizon scalability under γ . Sweeping $\gamma \in \{0.5, \dots, 0.99\}$ (Figure 6), unpruned propagation explodes ($78 \rightarrow 67,366$ knots over 12 backups at $\gamma = 0.9$), whereas DP pruning ($\epsilon_{\text{prune}} = 10^{-4}$) saturates knot counts (max 181/state). Exact-CPWL sustains a $28\text{--}264\times$ speedup

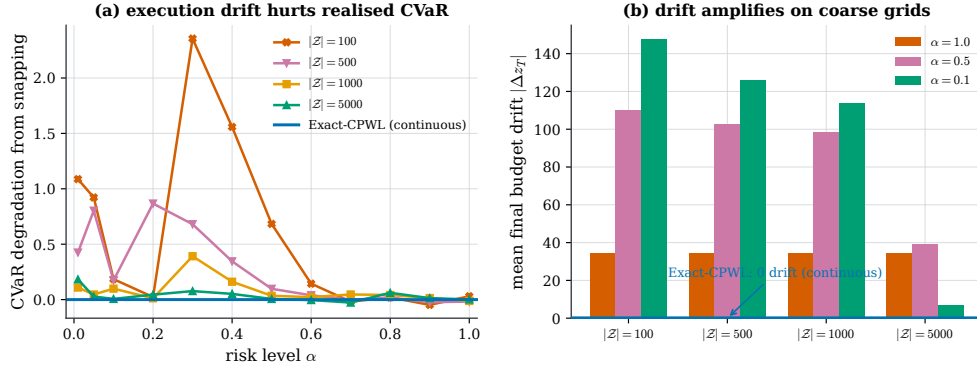


Figure 5: Snap-to-grid execution drift. (a) Realized-CVaR degradation from snapping grows on coarse grids and peaks in the mid-risk regime; Exact-CPWL (continuous) has zero degradation. (b) Mean final budget drift $|\Delta z_T|$ over 20 steps amplifies as the grid coarsens, reaching 97.5 at $|Z| = 100$.

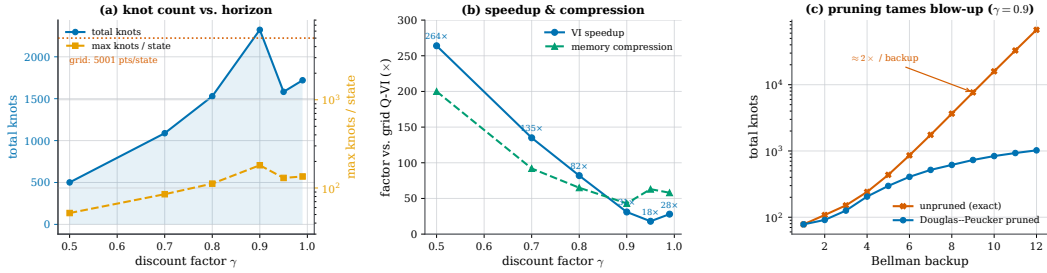


Figure 6: Horizon scalability under discount factor γ . (a) Total and per-state knot counts stay orders of magnitude below the fixed 5001-point grid. (b) Value-iteration speedup and memory compression over Grid Q-VI. (c) Douglas-Peucker pruning tames the $\approx 2\times$ -per-backup knot blow-up (log scale, $\gamma = 0.9$).

and 43–200 $\times$ compression over grid VI across horizons, retaining a 28 $\times$ speedup even at $\gamma = 0.99$ (effective horizon 100).

5 CONCLUSION

We introduced **Exact-CPWL CVaR**, a grid-free dynamic programming framework for static CVaR optimization in MDPs. By proving that the static CVaR Bellman operator strictly preserves 1D CPWL structure—even though $\max_a$ breaks concavity—Exact-CPWL eliminates spatial discretization, $O(\Delta/\alpha)$ tail-error amplification, and snap-to-grid drift. Through dynamic kink propagation, exact upper-envelope merging, bounded sup-norm DP pruning, and closed-form $O(|K_{s_0}|)$ knot extraction, it achieves a 35 $\times$ speedup, 43–70 $\times$ compression, and certified sup-norm bounds across three discrete benchmarks ($|\mathcal{S}| \leq 100$, $F \leq 16$); an $O(|\mathcal{A}||K| \log |\mathcal{A}|)$ divide-and-conquer merge further accelerates the envelope step by up to 1,449 $\times$. Beyond discrete states, we validate transfer to continuous nominal spaces via simplicial meshing (MountainCarCVaR-v0, $|\mathcal{S}| = 900$), retaining 10 $\times$ compression and drift-free 100% goal reach where the grid collapses. Because convex combinations and neural parameterizations preserve 1D CPWL structure in z , the framework extends further via neural knot networks and sample-based point-based DP (Appendix C)—a promising direction for scaling to high-dimensional control.

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A MATHEMATICAL PROOFS

A.1 PROOF OF PROPOSITION 3.1 (CPWL CLOSURE UNDER $\bar{T}$)

Let $\mathcal{F}_{\text{CPWL}}$ be the space of bounded CPWL functions on $[-r_\gamma, r_\gamma]$. For $\bar{v} \in \mathcal{F}_{\text{CPWL}}$,

$$Q(s, z, a) = z^- - (r(s, a) + z)_- + \gamma \sum_{s' \in \mathcal{S}} P(s'|s, a) \bar{v}\left(s', \text{clip}\left(\frac{r(s, a) + z}{\gamma}, -r_\gamma, r_\gamma\right)\right). \quad (9)$$

The per-step reward $\tilde{r}(s, z, a) = z^- - (r(s, a) + z)_-$ is CPWL with primary kinks at $z = 0$ and $z = -r(s, a)$. For each s' , $\bar{v}(s', \cdot)$ is linear on sub-intervals defined by $K_{s'}$; the inverse-affine map $z \mapsto (r(s, a) + z)/\gamma$ sends knots $z' \in K_{s'}$ to $z = \gamma z' - r(s, a)$. As linear transformations and linear combinations preserve CPWL structure, $Q(s, z, a)$ is CPWL with knot set $K_{s, a}$ (Eq. 5). Finally, $\bar{v}_{\text{new}}(s, z) = \max_a Q(s, z, a)$ is the pointwise maximum of finitely many CPWL functions; its upper envelope is CPWL with new vertices only at segment intersections where $\arg \max_a$ switches. Hence $\bar{T}\bar{v} \in \mathcal{F}_{\text{CPWL}}$. $\square$

A.2 NON-CONCAVITY VIA THE DISCRETE ACTION MAXIMUM

Let $f_1(z) = a_1 z + b_1$ and $f_2(z) = a_2 z + b_2$ be affine on $[z_A, z_B]$ with $a_1 < a_2$, intersecting at $z_0 \in (z_A, z_B)$. Then $g(z) = \max(f_1, f_2)$ equals f_1 for $z < z_0$ and f_2 for $z > z_0$, so its distributional second derivative at z_0 is a positive impulse $(a_2 - a_1) \delta(z - z_0) > 0$: g is locally convex at z_0 . Thus $\max_a Q(s, z, a)$ introduces locally convex action-switch vertices, and $\bar{T}$ destroys global concavity. This is consistent with the 6,699 measured concavity violations on the high-precision reference. $\square$

A.3 PROOF OF THEOREM 3.2 (CERTIFIED ERROR FLOOR)

Let $\bar{T}_{\text{DP}} = \Pi_{\text{DP}} \circ \bar{T}$, where $\|\Pi_{\text{DP}}(\bar{v}) - \bar{v}\|_\infty \leq \epsilon_{\text{prune}}$ for all $\bar{v}$. Since $\bar{T}$ is a γ -contraction,

$$\begin{aligned} \|\bar{v}_{\text{CPWL}} - \bar{v}^*\|_\infty &= \|\bar{T}_{\text{DP}}\bar{v}_{\text{CPWL}} - \bar{T}\bar{v}^*\|_\infty \\ &\leq \|\bar{T}_{\text{DP}}\bar{v}_{\text{CPWL}} - \bar{T}\bar{v}_{\text{CPWL}}\|_\infty + \|\bar{T}\bar{v}_{\text{CPWL}} - \bar{T}\bar{v}^*\|_\infty \\ &\leq \epsilon_{\text{prune}} + \gamma \|\bar{v}_{\text{CPWL}} - \bar{v}^*\|_\infty. \end{aligned} \quad (10)$$

Rearranging, $(1 - \gamma) \|\bar{v}_{\text{CPWL}} - \bar{v}^*\|_\infty \leq \epsilon_{\text{prune}}$, i.e., $\|\bar{v}_{\text{CPWL}} - \bar{v}^*\|_\infty \leq \epsilon_{\text{prune}} / (1 - \gamma)$. $\square$

A.4 PROOF OF THEOREM 3.3 (EXTREMUM AT KNOT VERTICES)

Let g be 1D CPWL on $[a, b]$ with ordered knots $\{z_1, \dots, z_N\}$ and $g(z) = m_j z + c_j$ on $I_j = [z_j, z_{j+1}]$. An affine function attains its interval maximum at an endpoint: $\max_{z \in I_j} g(z) = \max(g(z_j), g(z_{j+1}))$. Taking the maximum over all j gives $\max_{z \in [a, b]} g(z) = \max_j g(z_j)$, so the global maximum occurs at a knot vertex. $\square$

B ADDITIONAL TABLES AND FIGURES

Table 4: Continuous nominal state space (MountainCarCVaR-v0, 30×30 simplicial mesh, $|\mathcal{S}| = 900$, $|\mathcal{A}| = 3$, $\gamma = 0.99$; Section 4.4). Both policies are evaluated in the true continuous dynamics over 100 Monte Carlo rollouts.

METRIC	EXACT-CPWL	GRID Q-VI ($ \mathcal{Z}_\Delta = 1001$)
VALUE FUNCTION STORAGE	86,906 KNOTS	900,900 GRID NODES
MEMORY COMPRESSION	10.4×	1.0×
MAX KNOTS / NODES PER CELL	187	1,001
CERTIFIED ERROR $\ \bar{v}_{\text{CPWL}} - \bar{v}^*\ _\infty$	≤ 0.0523	UNCERTIFIED (OUTER GAP ≤ 7.09)
EMPIRICAL CVAR ($\alpha=0.01$)	-89.79 (100% GOAL REACH)	-98.21 (0% REACH, BOUNDARY TRAP)

Table 5: Closed-form outer static CVaR estimate Ψ^* and optimal budget z^* from Exact-CPWL versus grid baselines, with grid approximation gaps $|\Psi - \Psi^*|$.

α	EXACT z^*	EXACT Ψ^*	GRID Q-VI Ψ	GRID Q-LEAR. Ψ	GRID GAP $ \Psi - \Psi^* $
1.00	3.439	-7.7119	-7.7119	-7.7344	0.0000
0.90	3.439	-8.1867	-8.2159	-8.2408	0.0292
0.80	3.439	-8.7802	-8.8304	-8.8584	0.0502
0.70	3.439	-9.5432	-9.5727	-9.6036	0.0295
0.60	7.941	-10.4352	-10.5165	-10.4566	0.0813
0.50	9.114	-10.8328	-10.9150	-10.9281	0.0822
0.40	9.997	-11.1040	-11.1357	-11.1551	0.0317
0.30	9.997	-11.4730	-11.5036	-11.5294	0.0306
0.20	10.222	-12.1850	-12.2394	-12.2782	0.0544
0.10	11.473	-13.6492	-13.7673	-13.8307	0.1181
0.05	13.166	-15.0480	-15.1448	-15.2269	0.0968
0.01	16.223	-18.1941	-18.2843	-18.3707	0.0902

Table 6: Ablation 1: pruning tolerance ϵ_{prune} sweep—runtime, knot compression, Bellman residual, empirical value error, and theoretical-bound compliance.

ϵ_{prune}	VI TIME (S)	TOTAL KNOTS	MAX KNOTS/STATE	RESIDUAL $\ \bar{T}\bar{v} - \bar{v}\ _\infty$	EMP. VALUE ERROR	BOUND $\frac{\epsilon}{1-\gamma}$	WITHIN BOUND?
0.1	0.19	139	10	2.62E-01	1.3650	1.00E+00	EXCEEDED (EARLY STOP)
0.01	0.90	307	25	2.53E-02	0.0782	1.00E-01	YES
0.001	3.18	734	66	2.42E-03	0.0061	1.00E-02	YES
0.0001	11.38	2,322	181	2.60E-04	0.00065	1.00E-03	YES
1E-05	29.87	5,982	485	8.79E-05	0.000031	1.00E-04	YES
1E-06	73.35	14,906	1366	8.98E-05	REFERENCE	1.00E-05	—

Table 7: Per-backup complexity profiling across representative fan-out F and action-size $|\mathcal{A}|$ configurations ($|\mathcal{S}| = 20$, $\gamma = 0.9$, $\epsilon_{\text{prune}} = 10^{-4}$). Times are per state backup in milliseconds. Envelope merging dominates but post-pruning $|K|$ stays tightly bounded despite pre-merge candidate blow-up.

F	$ \mathcal{A} $	VI TIME (S)	BACKUP (MS)	CAND. GEN (MS)	MERGE (MS)	DP PRUNE (MS)	MERGE %	MAX CAND. KNOTS	POST-PRUNE $ K $
1	4	0.07	0.34	0.11	0.18	0.05	53%	28	4
2	4	2.88	3.98	0.36	2.81	0.80	71%	574	82
4	4	11.11	10.68	0.87	8.27	1.54	77%	1,403	132
8	4	29.21	29.88	2.28	24.98	2.62	84%	4,224	209
16	4	18.37	18.81	2.75	14.30	1.77	76%	2,702	136
16	16	8.62	11.69	3.51	7.83	0.36	67%	637	38

Table 8: Ablation 2: 1D curve-simplification operators across knot budgets K_{max} . Metrics are sup-norm value error $\|V - V_{\text{ref}}\|_\infty$ and mean CVaR estimation error.

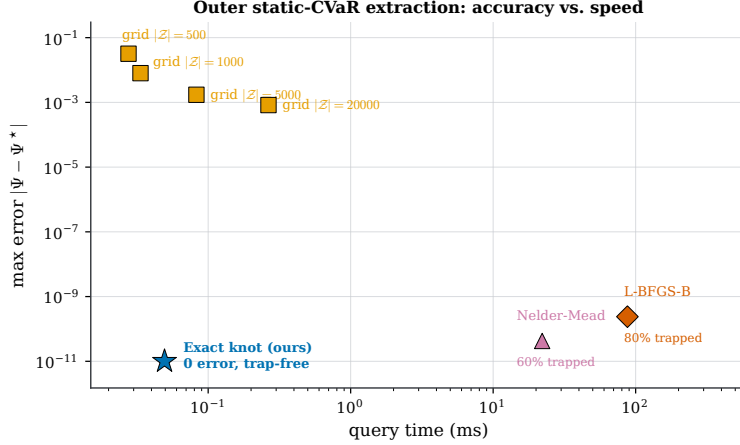
STRATEGY	K_{max}	BELLMAN RESIDUAL	SUP-NORM ERROR	MEAN CVaR ERROR	UNIFORM/DP GAP
DOUGLAS-PEUCKER (DP)	20	1.10E-02	2.072E-02	8.285E-02	1.0×
VISVALINGAM-WHYATT (VW)	20	3.34E-02	3.336E-02	4.312E-02	—
SLOPE-CHANGE	20	2.38E-01	3.989E-01	3.391E+00	—
UNIFORM SUBSAMPLING	20	9.45E+00	1.373E+01	2.492E+00	662.7× WORSE
DOUGLAS-PEUCKER (DP)	100	5.20E-04	9.209E-04	5.547E-03	1.0×
VISVALINGAM-WHYATT (VW)	100	8.10E-04	1.044E-03	1.139E-03	—
SLOPE-CHANGE	100	7.35E-02	1.196E-01	7.987E-01	—
UNIFORM SUBSAMPLING	100	6.23E+00	6.922E+00	3.420E-01	7516.2× WORSE

C EXTENSION TO CONTINUOUS NOMINAL STATE SPACES

While exact discrete-state dynamic programming provides certified convergence guarantees, the CPWL machinery of Exact-CPWL extends to continuous nominal state spaces $\mathcal{S} \subseteq \mathbb{R}^d$ through function approximation and meshing. The key enabling fact is that *all three* paradigms below preserve the 1D CPWL structure in the budget coordinate z , so closed-form $O(|K_{s_0}|)$ outer static-CVaR extraction (Theorem 3.3) and drift-free continuous rollout remain directly applicable.

Table 9: Ablation 3: outer static CVaR extraction—closed-form knot evaluation vs. continuous optimization.

EXTRACTION STRATEGY	QUERY TIME (MS)	MAX $ \Psi - \Psi^* $	TRAPS / 10 STARTS	TRAP RISK
EXACT KNOT EVAL (OURS)	0.0494	0.00E+00	0.0 / 10	0.0 %
GRID SWEEP ($ \mathcal{Z} = 1000$)	0.0335	8.00E-03	—	DISCRETIZATION LAG
GRID SWEEP ($ \mathcal{Z} = 20000$)	0.2650	8.26E-04	—	DISCRETIZATION LAG
L-BFGS-B (10 RESTARTS)	87.5600	2.39E-10	0.8 / 10	80.0% AT $\alpha = 0.60$
NELDER-MEAD (10 RESTARTS)	22.0773	4.25E-11	0.5 / 10	60.0% AT $\alpha = 0.60$

Figure 7: Ablation 3: accuracy vs. speed of outer static-CVaR extraction. Exact knot evaluation is exact and trap-free at sub-0.05 ms; dense grid sweeps trade time for residual discretization error, and gradient/simplex optimizers are orders of magnitude slower with high local-trapping rates at $\alpha = 0.60$.Table 10: Divide-and-conquer (D&C) sweep-line vs. pairwise-baseline upper-envelope merge for a single merge of $|\mathcal{A}|$ CPWL actions ($|K| = 100$ knots each). The two are numerically equivalent to machine precision; D&C scales as $\mathcal{O}(|\mathcal{A}| \log |\mathcal{A}|)$ while the baseline scales as $\mathcal{O}(|\mathcal{A}|^2)$.

$ \mathcal{A} $	BASILINE MERGE (MS)	D&C MERGE (MS)	SPEEDUP	SUP-NORM MAX DIFF.
10	36.46	3.71	9.8×	5.5E-13
50	3,822.18	29.30	130.4×	1.6E-13
100	30,256.07	71.61	422.5×	2.9E-11
200	241,620.36	166.79	1,448.6×	3.2E-11

Table 11: Ablation 4b: continuous-interpolation grid rollout baselines (snap, interp, interp_oz) vs. Exact-CPWL (cpwl) over 10 seeds $\times$ 10,000 Monte Carlo rollouts. Interpolation removes terminal drift $|\Delta z_T|$, but Exact-CPWL retains superior empirical CVaR and goal-reach.

BENCHMARK (α)	CPWL CVAR	SNAP	INTERP	INTERP_OZ	DRIFT $ \Delta z_T $ (SNAP/INTERP)	GOAL (CPWL/INTERP)
CLIFFWALK ($\alpha=0.50$)	-10.82 $\pm$ 0.02	-10.88 $\pm$ 0.02	-10.89 $\pm$ 0.02	-10.85 $\pm$ 0.04	81.51 / 0.00	100% / 30%
GRIDWORLD100 ($\alpha=0.30$)	-18.18 $\pm$ 0.05	-20.00 $\pm$ 0.00	-20.02 $\pm$ 0.00	-18.60 $\pm$ 0.07	186.35 / 0.00	100% / 1%
GRIDWORLD100 ($\alpha=0.10$)	-19.49 $\pm$ 0.03	-20.00 $\pm$ 0.00	-20.05 $\pm$ 0.00	-19.95 $\pm$ 0.08	186.13 / 0.00	100% / 1%

C.1 SIMPLICIAL INTERPOLATION OVER NOMINAL SPACE (MESHING)

Discretize $\mathcal{S}$ with a simplicial mesh (e.g., Delaunay triangulation) with vertex set V_S . At each vertex $v \in V_S$ store a 1D CPWL function $\bar{v}(v, z)$ with knot set K_v . For an arbitrary continuous state s in the enclosing simplex $\Delta(v_0, \dots, v_d)$, evaluate by barycentric interpolation,

$$\bar{v}(s, z) = \sum_{i=0}^d \lambda_i(s) \bar{v}(v_i, z), \quad \sum_{i=0}^d \lambda_i(s) = 1, \quad \lambda_i(s) \geq 0. \quad (11)$$

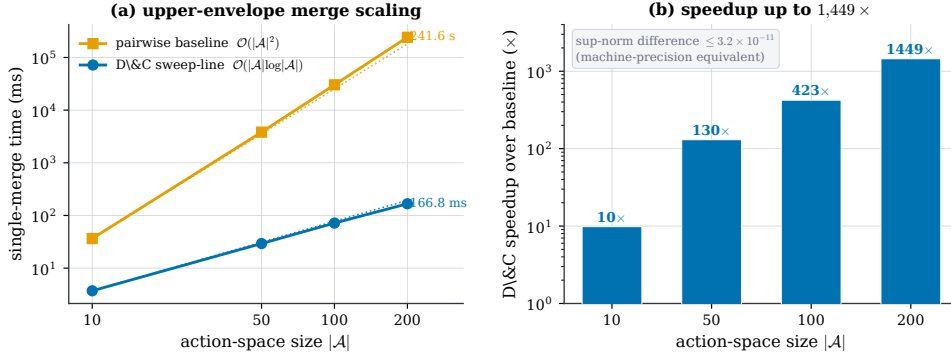


Figure 8: Divide-and-conquer (D&C) sweep-line upper-envelope merging vs. the pairwise baseline ($|K|=100$ knots per action; Section 4.5). **(a)** Single-merge time (log-log): the baseline scales as $|\mathcal{A}|^{2.84}$ while D&C scales as $|\mathcal{A}|^{1.32}$ (dotted fits). **(b)** D&C speedup grows from $10\times$ at $|\mathcal{A}|=10$ to $1,449\times$ at $|\mathcal{A}|=200$; both algorithms produce machine-precision-identical envelopes ($\leq 3.2 \times 10^{-11}$).

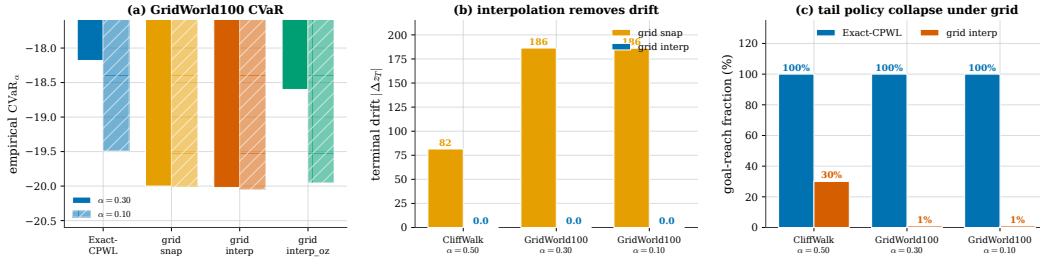


Figure 9: Continuous-interpolation grid rollout baselines (snap, interp, interp_oz) vs. Exact-CPWL (Ablation 4b). **(a)** On GridWorld100, drift-free interp still trails Exact-CPWL in empirical CVaR $_{\alpha}$. **(b)** Interpolation drives terminal budget drift $|\Delta z_T|$ to zero, matching Exact-CPWL. **(c)** At tight risk, grid value error collapses the interpolated policy’s goal-reach fraction to $\sim 1\%$, while Exact-CPWL reaches the goal 100% of the time.

Because a convex combination of 1D CPWL functions in z is exactly 1D CPWL, $\bar{v}(s, \cdot)$ remains CPWL for every $s \in \mathcal{S}$, preserving closed-form outer optimization and drift-free execution. This paradigm is empirically validated on MountainCarCVaR-v0 in Section 4.4, where a 30×30 multilinear mesh yields $10.4\times$ memory compression and 100% goal reach against a 3D grid baseline.

C.2 NEURAL KNOT NETWORK / IMPLICIT CPWL APPROXIMATION

For high-dimensional $\mathcal{S}$, replace tabular state arrays with a parametric network $K_{\phi}(s)$ that outputs knot locations and values conditioned on s :

$$K_{\phi}(s) = \{(z_i(s; \phi), v_i(s; \phi))\}_{i=1}^M. \quad (12)$$

Equivalently, Maxout or continuous piecewise-linear (CPWL) layers parameterized over $\mathcal{S} \times [-r_{\gamma}, r_{\gamma}]$ let the nominal state s condition the slope/intercept vectors while constraining the output in z to be CPWL. Bellman updates minimize the projected sup-norm Bellman error

$$\mathcal{L}(\phi) = \mathbb{E}_{s \sim \mathcal{D}} [\|\bar{v}_{\phi}(s, \cdot) - \bar{T}_{\text{DP}} \bar{v}_{\phi_{\text{target}}}(s, \cdot)\|_{\infty}]. \quad (13)$$

Since the output is CPWL in z , closed-form knot evaluation for outer static-CVaR optimization remains applicable.

C.3 SAMPLE-BASED POINT-BASED DYNAMIC PROGRAMMING

Analogous to point-based value iteration (PBVI) in POMDPs, sample a representative set $\mathcal{S}_{\text{sample}} \subset \mathcal{S}$, perform exact 1D CPWL Bellman backups (in z) at those points, and interpolate values at unvisited states via kernel or nearest-neighbor representations. This trades the certified discrete-state guarantee for scalability while retaining exact per-point CPWL structure.

D HYPERPARAMETERS AND REPRODUCIBILITY

Table 12 lists all settings for reproducing the `CliffWalkCVaR-v0` experiments.

Table 12: Complete hyperparameter specification for Exact-CPWL and baselines.

HYPERPARAMETER	VALUE
ENVIRONMENT DIMENSIONS	4×5 GRIDWORLD
STEP PENALTY	-1
CRATER ENTRY PENALTY	-10
GOAL REWARD	0 (ABSORBING)
SLIP PROBABILITY ω	0.25
DISCOUNT FACTOR γ	0.90
BUDGET DOMAIN $[z_{\min}, z_{\max}]$	$[-110.0, 110.0]$
EXACT-CPWL PRUNING TOLERANCE ϵ_{PRUNE}	1.0×10^{-4}
EXACT-CPWL CONVERGENCE TOLERANCE δ	1.0×10^{-4}
GRID VI / Q-LEARNING BINS $ \mathcal{Z}_{\Delta} $	5,001
Q-LEARNING TRAINING EPISODES M	75,000
Q-LEARNING EXPLORATION SCHEDULE ϵ	LINEAR DECAY $1.0 \rightarrow 0.1$
Q-LEARNING LEARNING-RATE PARAMETERS	$\kappa = 1.0, \kappa_{\min} = 10^{-4}, \lambda = 0.01$
MONTE CARLO ROLLOUTS PER α	10,000
INDEPENDENT RANDOM SEEDS	10